Online Examination System

A Minor Project Report

Submitted in partial fulfillment of requirement of the

Degree of

BACHELOR OF TECHNOLOGY in COMPUTER SCIENCE & ENGINEERING

BY

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Report Approval

The project work **“Online Examination System”**is hereby approved as a creditable study of an engineering/computer application subject carried out and presented in a manner satisfactory to warrant its acceptance as prerequisite for the Degree for which it has been submitted.

It is to be understood that by this approval the undersigned do not endorse or approve any statement made, opinion expressed, or conclusion drawn there in; but approve the “Project Report” only for the purpose for which it has been submitted.

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I/We hereby declare that the project entitled **"Online Examination System"** submitted in partial fulfillment for the award of the degree of Bachelor of Technology in **'Department of Computer Science and Technology'** completed under the supervision of **Prof. Vivek Kumar Gupta, Assistant Professor, Department Computer Science and Engineering,** Faculty of Engineering, Medi-Caps University Indore is an authentic work.

Further, I/we declare that the content of this Project work, in full or in parts, have neither been taken from any other source nor have been submitted to any other Institute or University for the award of any degree or diploma.

Signature and name of the student(s) with date

Madhyama Mandloi

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Certificate

I/We, **Prof. Vivek Kumar Gupta** certify that the project entitled **"Online Examination System"**submitted in partial fulfillment for the award of the degree of Bachelor of Technology by **Madhyama Mandloi, Mahi Rathore and Atharv Goutam** is the record carried out by them under my/our guidance and that the work has not formed the basis of award of any other degree elsewhere.

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Abstract

In the modern era of digital transformation, automated examination systems play a vital role in improving efficiency and accessibility in education. The **Online Examination System** is a web-based application designed to conduct and manage exams digitally using a client-server architecture with HTML, CSS, JavaScript, Java Servlets, and MySQL.

The system allows students to register, log in, attempt multiple-choice exams, and view results instantly. Administrators can create exams, manage questions, and monitor performance through a centralized dashboard. Secure authentication ensures controlled access and data integrity.

With an intuitive interface and automated evaluation, the system reduces manual effort, minimizes errors, and provides quick and accurate results, making it a reliable solution for modern online assessments.

Keywords:

Keyword	**Explanation**
HTML	Used to structure the web pages of the system.
CSS	Used to design and style the user interface.
JavaScript	Adds interactivity and handles client-side logic.
Java Servlets	Handles server-side processing and request handling.
MySQL	Stores user data, exam details, and results.
Authentication	Ensures secure login and access control.
Online Examination	Conducting exams through a web-based platform.
Automated Evaluation	Automatically calculates and displays results.

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Abbreviations

- **UI –** User Interface
- **DB –** Database
- **HTML –** Hyper Text Markup Language
- **CSS –** Cascading Style Sheets
- **JS –** JavaScript
- **JVM –** Java Virtual Machine
- **JDBC –** Java Database Connectivity
- **SQL –** Structured Query Language
- **HTTP –** Hypertext Transfer Protocol
- **URL –** Uniform Resource Locator
- **URI –** Uniform Resource Identifier
- **API –** Application Programming Interface
- **MVC –** Model View Controller
- **JSP –** Java Server Pages
- **DAO –** Data Access Object
- **CRUD –** Create, Read, Update, Delete
- **IDE –** Integrated Development Environment
- **OS –** Operating System
- **RAM –** Random Access Memory
- **CPU –** Central Processing Unit
- **AJAX –** Asynchronous JavaScript and XML
- **JSON –** JavaScript Object Notation
- **Session –** User session for maintaining login state
- **Authentication –** User identity verification process
- **Authorization –** Access control mechanism
- **Tomcat –** Apache Tomcat Web Server
- **MySQL –** Relational Database Management System5

These abbreviations represent the key technical terms, configuration keys, protocols, libraries, and variable short forms used throughout the Online Examination System.

Notations & Symbols

Symbol / Notation	Meaning / Purpose in Online Examination System
+	Addition operator (used in score calculation)
-	Subtraction operator (used in marks deduction if applicable)
*	Multiplication operator
/	Division operator
%	Modulo operator (used in logic operations)
== / ===	Equality comparison operators
!= / !==	Not equal comparison operators
<	Less-than operator
>	Greater-than operator
!	Logical NOT operator
&&	Logical AND operator
||	Logical OR operator
? :	Ternary conditional operator
()	Used for grouping conditions and function calls
{ }	Block of code in programming
[ ]	Arrays and list structures
.	Object property access (dot notation)
_	Used in variable naming (e.g., user_id)
$	Used in strings or UI display
/	URL path separator
session	Represents user login session
request (req)	Client request object
response (res)	Server response object
id	Unique identifier for database records
localhost	Local server address
port	Server communication endpoint

Chapter 1: Introduction

1.1 Introduction

In the modern educational landscape, the transition from traditional paper-based testing to digital platforms has become a necessity for academic institutions and corporate training centers. Manual examination processes are often plagued by logistical challenges, including paper wastage, manual grading errors, and the time-consuming nature of physical distribution.

The Online Examination System is a web-based application designed to automate the entire examination process, from test creation and scheduling to automated grading and performance analytics. This system provides a secure and efficient environment where students can take exams remotely and instructors can manage assessments with ease. By leveraging a robust tech stack, the system ensures data integrity and real-time feedback for users.

1.2 Literature Review

Existing research in educational technology emphasizes the shift toward automated assessment tools to improve scalability and objectivity. Studies on web-based examination systems highlight the importance of the Model-View-Controller (MVC) architecture in maintaining a clean separation between business logic and user interface, which enhances system maintainability.

Previous implementations have shown that using Java (Servlets & JSP) provides a secure, server-side environment suitable for handling concurrent user sessions during peak exam hours. Furthermore, the integration of data visualization libraries, such as Chart.js, has been identified as a key factor in helping administrators interpret student performance trends through graphical dashboards.

1.3 Objectives

Main Objective

To design and develop a secure, scalable, and user-friendly **Online Examination System** that automates the process of conducting exams and evaluating results digitally.

Supporting Objectives

• To develop a secure login and authentication system for students and administrators
• To enable administrators to create, manage, and schedule examinations
• To design a user-friendly interface for students to attempt exams

• To implement a timed examination system with multiple-choice questions
• To automate the evaluation process for instant result generation
• To maintain a database for storing user details, exam data, and results
• To ensure data security, reliability, and system performance

1.4 Significance

The Online Examination System enhances the traditional examination process by converting it into a digital and automated platform. It reduces manual effort, saves time, and minimizes errors through automated evaluation and instant result generation.

The system improves accessibility by allowing students to take exams from any location and ensures efficient data management through secure storage of user and exam information.

Overall, it increases efficiency, accuracy, and reliability in conducting examinations.

1.5 Research Design

The Online Examination System is developed using a structured approach based on **MVC architecture**, where the frontend (HTML, CSS, JavaScript) handles the interface, the backend (Java Servlets) manages logic, and MySQL is used for data storage.

The design process includes requirement analysis, system and database design, frontend and backend development, and testing for performance and security.

This approach ensures the system is scalable, secure, and efficient.

1.6 Source of Data

The Online Examination System utilizes various sources of data for its operation and functionality. These include:

• **User Input Data**: Information provided by students and administrators such as login credentials, exam responses, and exam details
• **Database Records**: Stored data including user information, question banks, exam schedules, and results maintained in MySQL database
• **System Generated Data**: Automatically generated results, scores, and performance reports

• **Reference Materials**: Academic research papers, online resources, and existing system studies used during system design and development

These data sources ensure proper functioning, accurate evaluation, and efficient management of the system.

1.7 Chapter Scheme

• Chapter 1: Introduction – Overview of the system
• Chapter 2: Requirement Specification – System requirements
• Chapter 3: System Design – Architecture and diagrams
• Chapter 4: Implementation, Testing & Maintenance – Development and testing
• Chapter 5: Results and Discussion – Outputs and UI
• Chapter 6: Summary and Conclusion – Overall findings
• Chapter 7: Future Scope – Enhancements

Chapter 2: Requirement Specification

2.1 User Characteristics

The primary users of the **Online Examination System** include:

• **Students** who want to appear for online examinations and view their results.
• **Administrators** who manage exams, questions, and student performance.
• **Educational institutions** that require a digital platform for conducting exams efficiently.

Users are expected to:

• Have basic knowledge of using web applications
• Be able to navigate through login, exam, and result interfaces
• Understand basic authentication processes (registration/login)

No advanced technical knowledge is required, as the system is designed to be simple, intuitive, and user-friendly.

2.2 Functional Requirements

The Online Examination System must perform the following core functions:

1. User Management

• Register new users with details such as name, email, and password
• Authenticate users through secure login
• Allow role-based access (Admin / Student)
• Maintain user session during exam

2. Exam Management (Admin)

• Create new exams with name and duration
• Add, edit, and delete questions
• Manage question banks
• Schedule exams

3. Exam Attempt (Student)

• Display available exams
• Allow students to start exams
• Show multiple-choice questions
• Provide timer-based exam interface
• Allow navigation between questions
• Submit exam

4. Result Generation

• Automatically evaluate answers
• Calculate score
• Display result (marks, pass/fail)
• Store results in database

5. System Features

• Secure login and authentication
• Timer-based exam system
• Error handling and validation
• Session management

2.3 Dependencies

The Online Examination System depends on:

• Internet connectivity for accessing the web application
• Web browser (Chrome, Edge, Firefox, etc.)
• Apache Tomcat server for backend execution
• MySQL database for storing data
• Java Servlets for backend processing
• HTML, CSS, and JavaScript for frontend

System performance may depend on server speed and network conditions.

2.4 Performance Requirements

• The system should respond quickly to user actions
• Login and exam loading should occur with minimal delay
• The system must support multiple users simultaneously
• Result generation should be fast and accurate
• Timer functionality should work reliably during exams

Performance is ensured through efficient database queries and optimized system design.

2.5 Hardware Requirements

Server Side

• Processor: Intel i5 or higher
• RAM: Minimum 8 GB
• Storage: Minimum 500 MB
• Operating System: Windows / Linux / macOS

Client Side

• Computer or Laptop
• Modern web browser
• Stable internet connection

2.6 Constraints & Assumptions

Constraints

• System requires continuous internet connectivity
• Performance depends on server capacity
• Limited to web-based platform
• Security depends on proper authentication mechanisms

Assumptions

• Users provide correct and valid information
• System is used in a secure environment
• Database is properly configured
• Users follow instructions during exams

Chapter 3: System Design

3.1 Algorithm (Online Examination Flow)

• User Registration/Login with authentication
• User selects available exam
• System validates user and exam access
• Questions are fetched from database
• Student attempts questions within time limit
• Answers are submitted
• System evaluates responses automatically
• Result is calculated and stored
• Result is displayed to the user

3.2 Function-Oriented Design

```
online-exam-system/
│
├── client/
│   ├── css/
│   │   └── style.css
│   ├── js/
│   │   └── script.js
│   ├── pages/
│   │   ├── login.html
│   │   ├── register.html
│   │   ├── dashboard.html
│   │   ├── exam.html
│   │   └── result.html
│   └── index.html
│
├── server/
│   ├── controller/
│   │   ├── LoginServlet.java
│   │   ├── RegisterServlet.java
│   │   ├── ExamServlet.java
│   │   └── ResultServlet.java
│   │
│   ├── dao/
│   │   ├── UserDAO.java
│   │   ├── ExamDAO.java
│   │   ├── QuestionDAO.java
│   │   └── ResultDAO.java
│   │
│   ├── model/
│   │   ├── User.java
│   │   ├── Exam.java
│   │   ├── Question.java
│   │   └── Result.java
│   │
│   ├── db/
│   │   └── DBConnection.java
│   │
│   └── web.xml
│
├── database/
│   └── schema.sql
│
```

Fig 3.0 File Structure Diagram

• **User Management Module** – Registration, login, and user handling
• **Authentication Module** – Login validation and session management
• **Exam Management Module** – Exam creation and scheduling (Admin)
• **Question Management Module** – Add, edit, delete questions
• **Result Module** – Evaluation and result generation
• **Frontend Module** – UI for login, exam interface, and results.

3.3 System Architecture

• **Frontend**: HTML, CSS, JavaScript (User Interface)
• **Backend**: Java Servlets (Business Logic & Request Handling)
• **Database**: MySQL (User, Exam, Question, Result storage)
• **Server**: Apache Tomcat

Architecture Type: Client–Server Architecture

3.4 Data Flow Design

Level 0 DFD:

User → Web Interface → Server → Database → Server → User

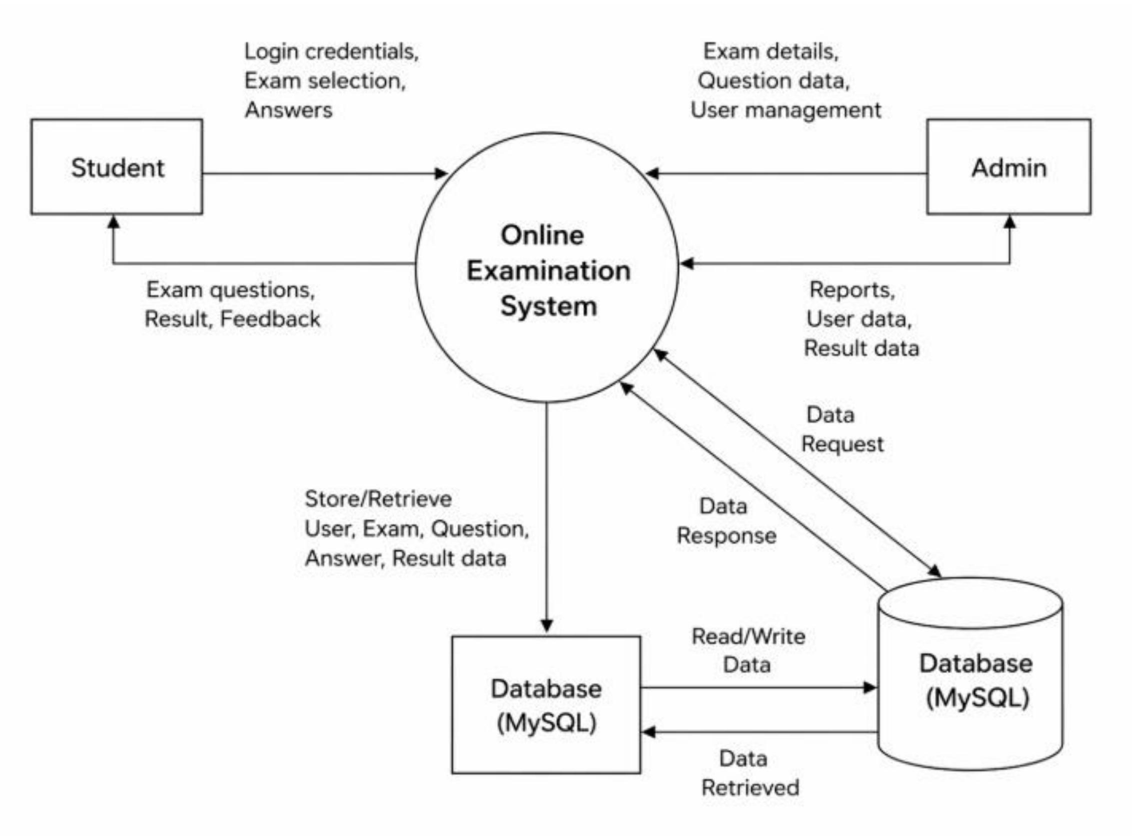


Fig 3.1 Data Flow Diagram Level 0

Level 1 DFD:

• Validate User Login
• Fetch Exam Questions
• Store Student Answers
• Evaluate Answers
• Generate Result

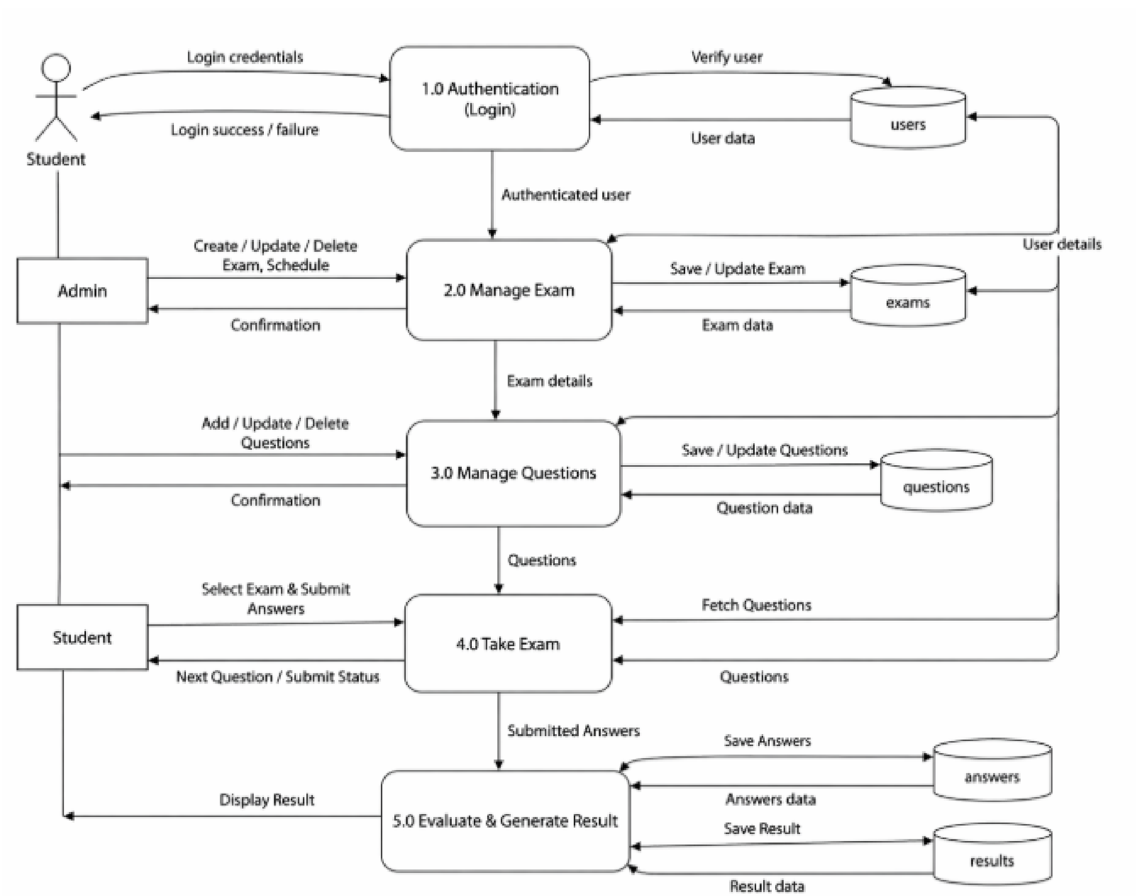


Fig 3.2 Data Flow Diagram Level 1

3.5 UML Design Components

Use Cases:

• Register
• Login
• Take Exam
• Submit Exam
• View Result
• Manage Exams (Admin)

Main Classes:

- User
- Student
- Admin
- Exam
- Question
- Result

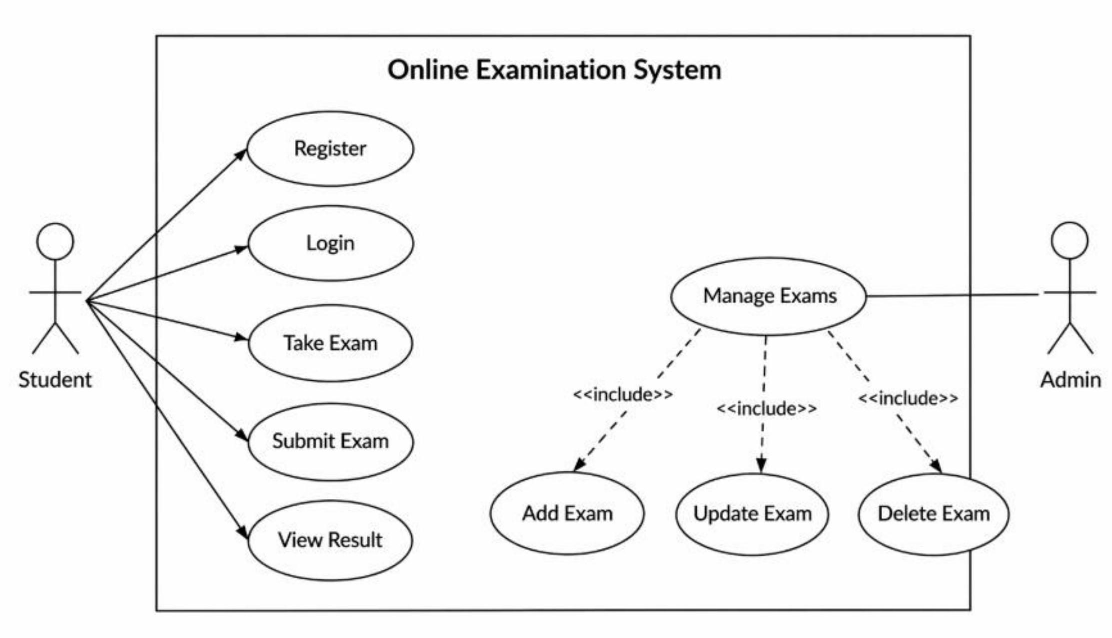


Fig 3.3 Use Case Diagram

Flowchart Flow:

User → Login/Register → Dashboard → Select Exam → Attempt Exam → Submit → Evaluate → Result → Logout

Steps:

Login/Register → Validate User → Access Dashboard → Select Exam → Start Exam → Answer Questions → Submit Exam → Evaluate Answers → Display Result

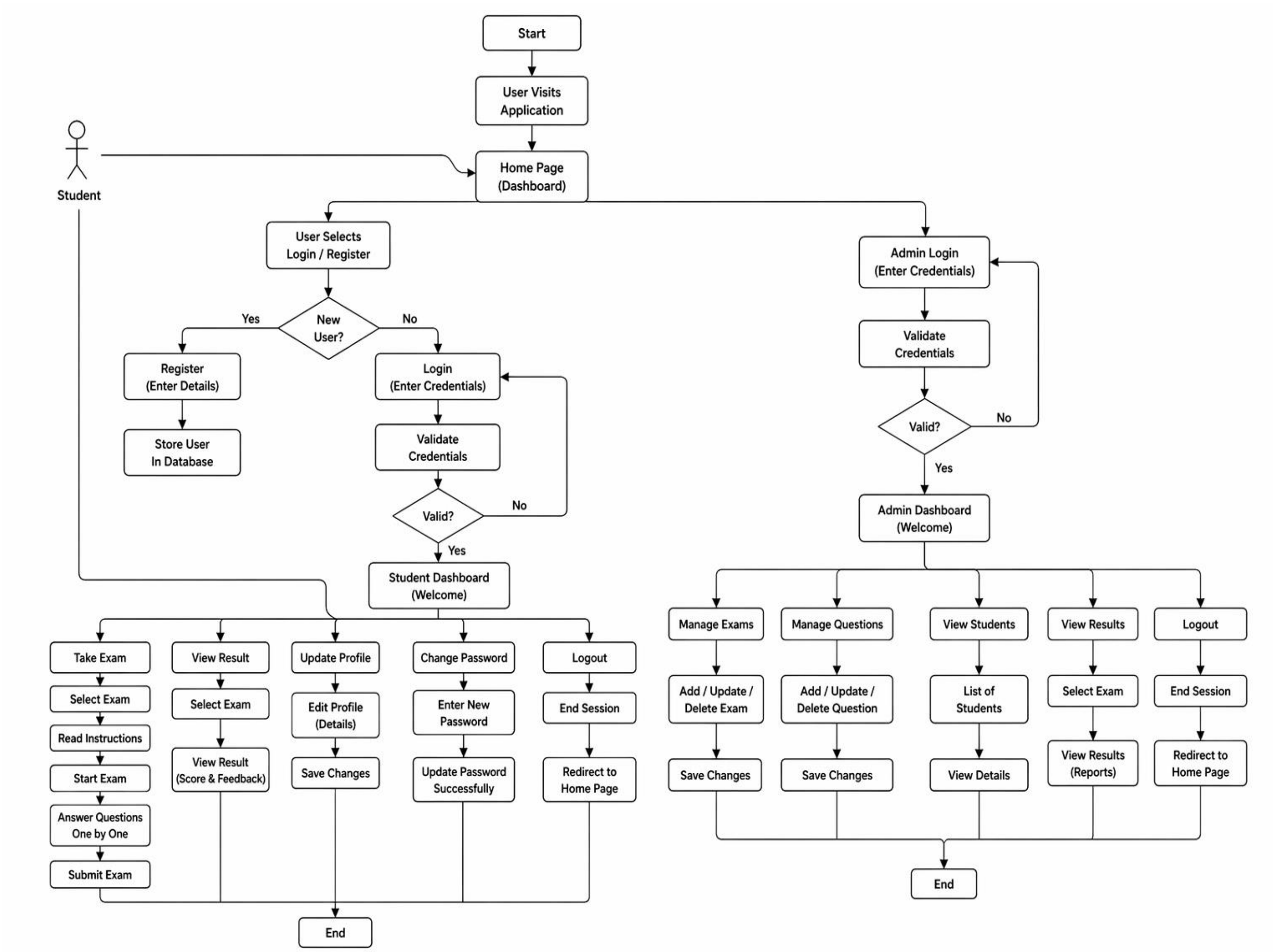


Fig 3.4 Flow Chart Diagram

Class Diagram Structure:

User → Student/Admin → Exam → Question → Result

Relationships:

User → (inherits) → Student, Admin
Student → attempts → Exam
Exam → contains → Question
Student → gets → Result
Result → belongs to → Exam

Main Classes & Functions:

User → login()
Student → takeExam(), viewResult()

Admin → createExam(), addQuestion()
Exam → manageExam()
Question → manageQuestion()
Result → calculateResult()

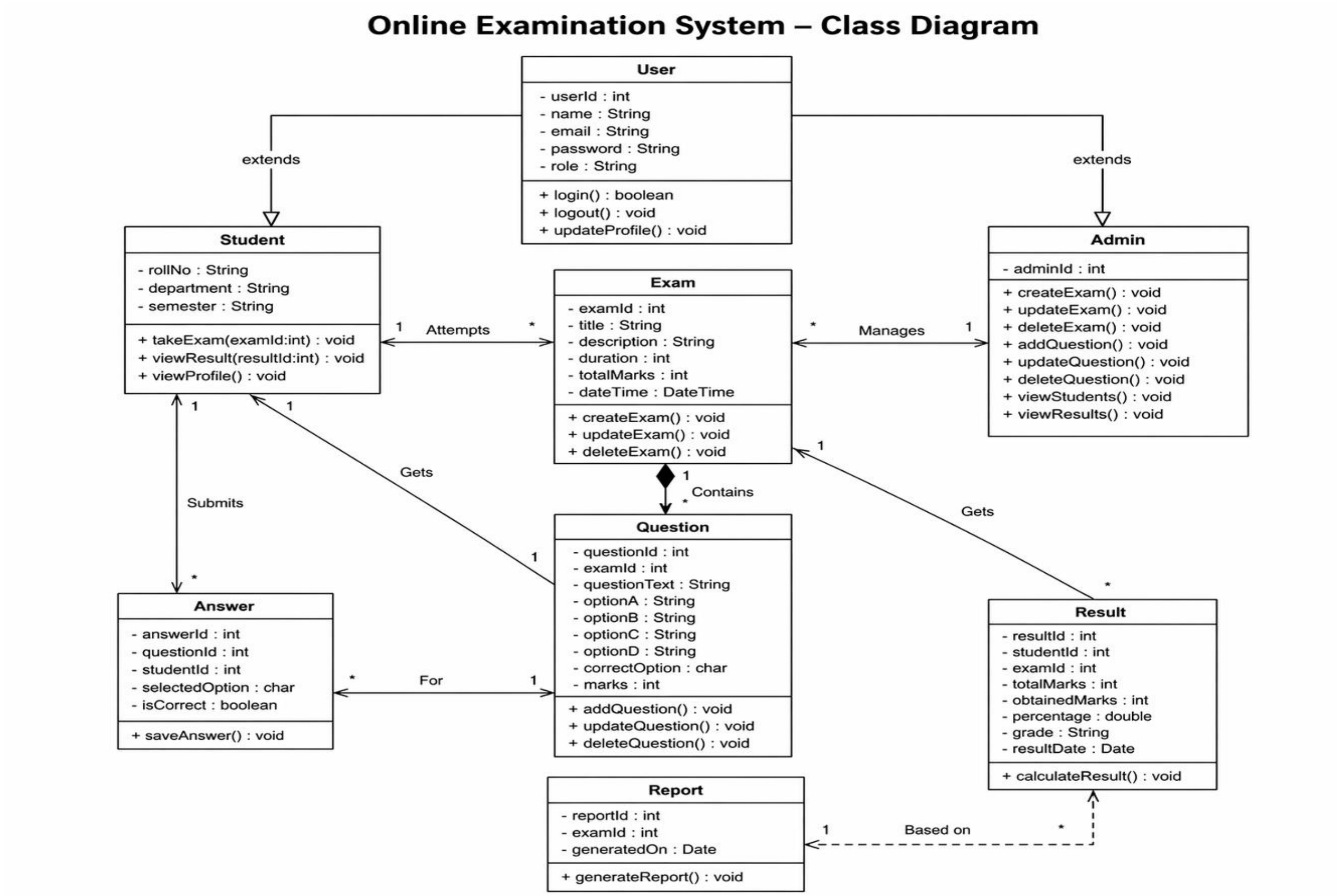


Fig 3.5 Class Diagram

Sequence Flow:

User → Frontend → Server → Database → Server → Frontend

Steps:
Login → Fetch Questions → Attempt Exam → Submit → Evaluate → Result

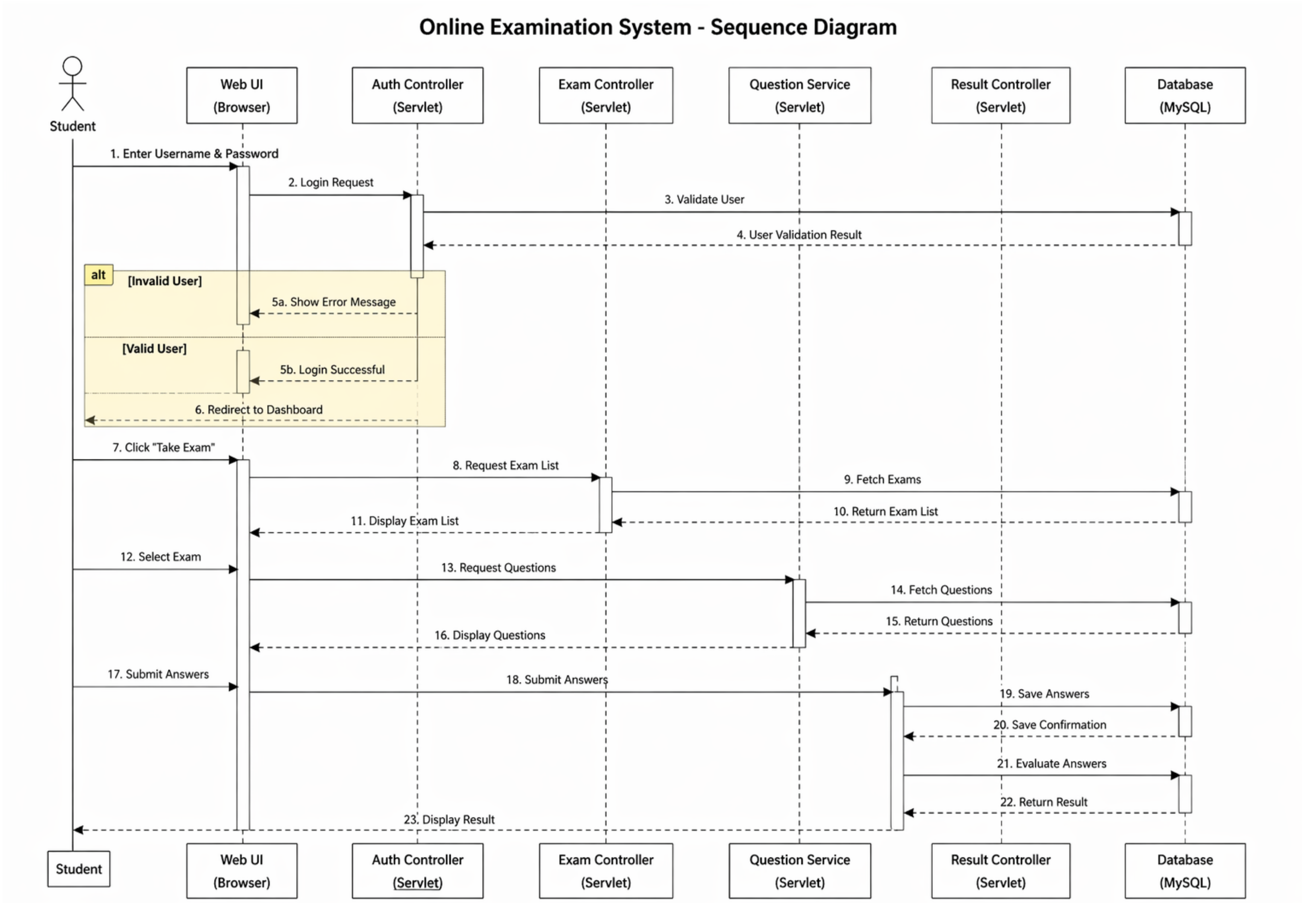


Fig 3.6 Sequence Diagram

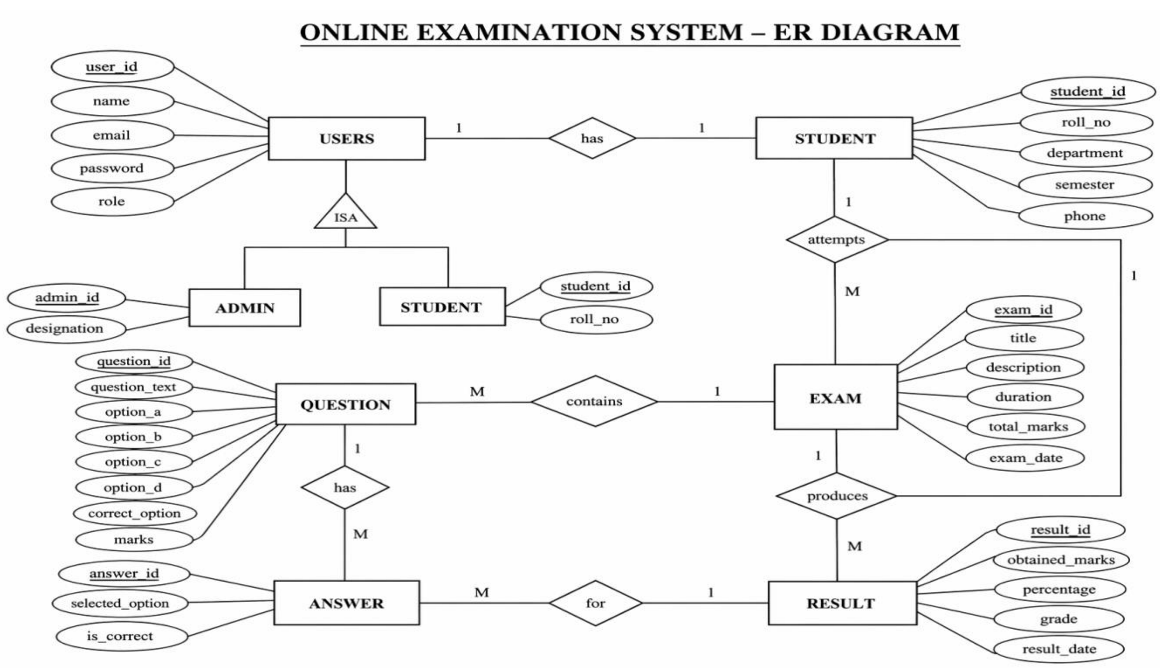


Fig 3.7 ER Diagram

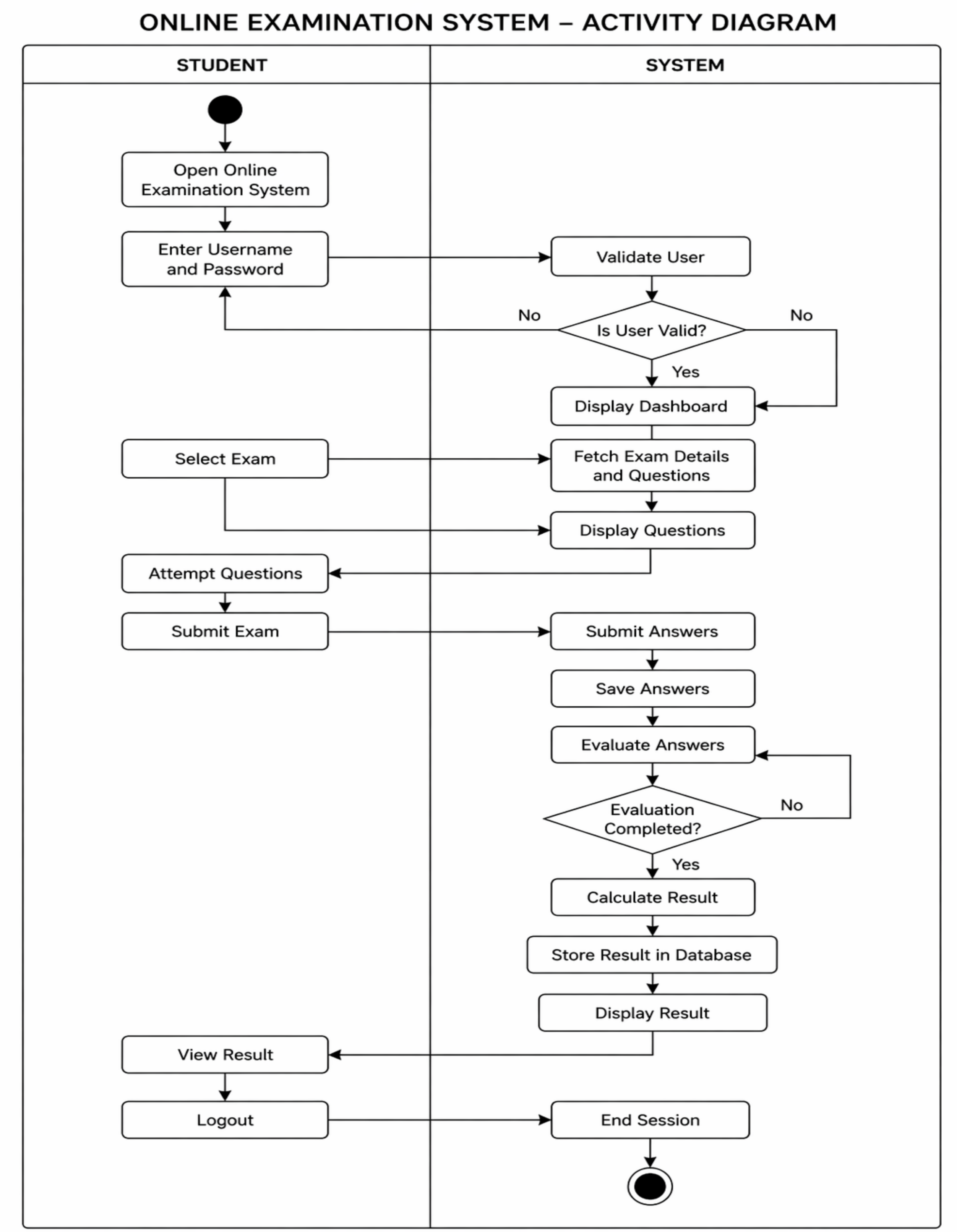


Fig 3.8 Activity Diagram

3.6 Non-Functional Design Considerations

• **Security**: User authentication and session management
• **Performance**: Fast response and efficient data handling
• **Scalability**: Supports multiple users simultaneously
• **Reliability**: Proper error handling and system stability

Conclusion

The system design of the Online Examination System follows a modular and scalable client-server architecture. It ensures secure authentication, efficient exam handling, and accurate result generation, providing a reliable platform for conducting online examinations.

Chapter 4: Implementation, Testing, and Maintenance

4.1 Languages, IDEs, Tools, and Technologies Used for Implementation

The **Online Examination System** is implemented using a client–server architecture, integrating frontend technologies, backend processing, and database management for efficient exam handling and result generation.

Frontend & Backend Technologies

• **Frontend**: HTML, CSS, JavaScript (for user interface and interaction)
• **Backend**: Java Servlets (for handling requests and business logic)
• **Server**: Apache Tomcat
• **Database**: MySQL

Programming Languages

• **Java** — used for backend development (Servlets)
• **HTML** — used for designing web pages
• **CSS** — used for styling user interface
• **JavaScript** — used for client-side validation and dynamic behavior

Styling & UI Tools

• CSS (for layout and design)
• Bootstrap (optional, for responsive design)
• JavaScript DOM manipulation (for interactivity)

Development Tools

• **Eclipse / IntelliJ IDEA / VS Code** — IDE for development
• **Apache Tomcat** — Web server for deployment
• **MySQL Workbench** — Database management
• **GitHub** — Version control

Database & Data Handling

• **JDBC (Java Database Connectivity)** — connects Java with MySQL
• **SQL Queries** — for data insertion, update, and retrieval
• **Servlet API** — for request-response handling

System Implementation Approach

The system is developed using a modular client–server approach where:

- The frontend handles user interaction (login, exam, result display)
- The backend processes logic (authentication, exam handling, evaluation)
- The database stores user data, questions, and results

This architecture ensures that the system is efficient, scalable, and easy to maintain.

4.2 Testing Techniques and Test Plans

The **Online Examination System** was tested using structured manual and integration testing methods to ensure proper functionality, accuracy, and reliability.

Unit Testing

- User registration and login functionality tested independently
- Input validation (email, password) verified
- Exam creation and question addition tested
- Timer functionality checked for accuracy
- Result calculation logic tested

Integration Testing

- Frontend-to-backend communication tested through HTTP requests
- Backend-to-database connectivity verified using JDBC
- Data retrieval (questions, results) tested
- Session management and authentication verified

User Testing

- Students tested login, exam attempt, and result viewing
- Admin tested exam creation, question management, and result monitoring
- UI usability and navigation tested
- Error messages validated for invalid login and incomplete submissions

Performance Testing

- System tested with multiple users simultaneously
- Exam loading and submission response time evaluated

• Result generation speed verified
• Server response under load tested

Conclusion

Testing of the system was mainly manual and integration-based. The system was verified to be reliable, efficient, and user-friendly for conducting online examinations.

4.3 Installation Instructions

Backend Setup

1. Install Java JDK (latest version).
2. Install Apache Tomcat Server.
3. Open the project in IDE (Eclipse / IntelliJ).
4. Configure database connection in DBConnection.java:

```
URL = jdbc:mysql://localhost:3306/online_exam
Username = root
Password = your_password
```

5. Start the Tomcat Server:

```
Run on Server (Tomcat)
```

Frontend Setup

1. Navigate to the project web directory.
2. Ensure all HTML, CSS, JS files are properly linked.
3. Deploy project on Tomcat server.
4. Open the application in browser:

```
http://localhost:8080/onlineexam
```

4.4 End User Instructions

1. Open the Online Examination System in a browser.
2. Register with name, email, and password.
3. Login using credentials.
4. Select an available exam from dashboard.
5. Click on Start Exam.
6. Answer all questions within given time.
7. Click on Submit Exam.
8. View result (score and status).

Chapter 5: Results and Discussions

5.1 User Interface Representation

The **Online Examination System** provides a clean, responsive, and user-friendly web interface developed using HTML, CSS, and JavaScript. The interface is designed to support both **students and administrators** efficiently.

1. Login Page

The Login page is the entry point of the system. It includes:

• Input fields for Email/Username and Password
• Role selection (Student/Admin)
• Login button
• “Forgot Password” option

This page ensures secure authentication before accessing the system.

2. Student Dashboard

After login, students are redirected to the dashboard. It includes:

• List of available exams
• “Start Exam” button
• Upcoming and completed exams
• Performance summary (scores, rank)

This page allows students to easily access and manage exams.

3. Examination Interface

This is the core functional interface of the system. It includes:

• Display of MCQ questions
• Multiple options for each question
• Timer countdown
• Navigation buttons (Next / Previous)
• Submit Exam button

Ensures smooth and timed exam experience.

4. Result Page

After submission, the result page displays:

• Total score
• Correct/incorrect answers
• Pass/Fail status

Provides instant feedback to students.

5. Admin Dashboard

Admin interface includes:

• Create Exam
• Add/Edit/Delete Questions
• View Students
• Monitor Results

Helps admin manage the entire system efficiently.

5.2 Brief Description of Various Modules of the System

The Online Examination System follows a modular approach:

Authentication Module

• User registration and login
• Validation of credentials
• Session management

Exam Management Module

• Create and manage exams
• Set duration and marks
• Schedule exams

Question Management Module

• Add, edit, delete questions
• Store MCQs with options and answers

Exam Attempt Module

• Display questions
• Capture student responses
• Timer-based submission

Result Module

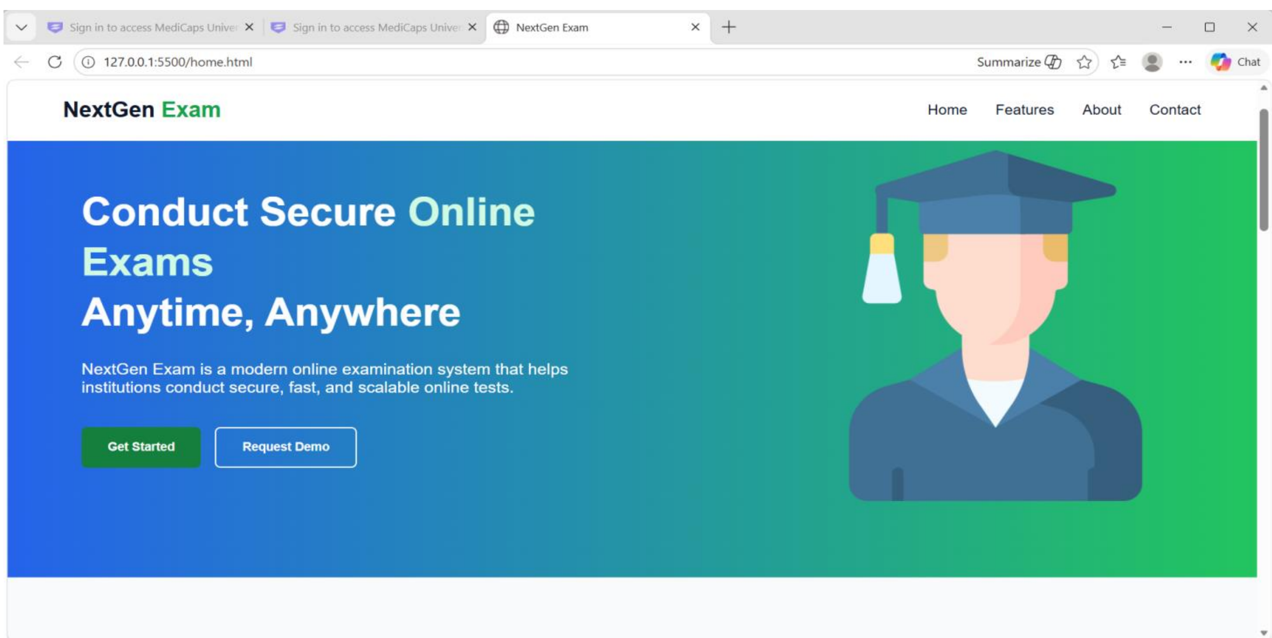
• Evaluate answers automatically
• Calculate score
• Display result instantly

Database Module

• Stores users, exams, questions, and results
• Ensures secure and efficient data handling

5.3 Snapshots of the System (Home Page)

The Home page presents:



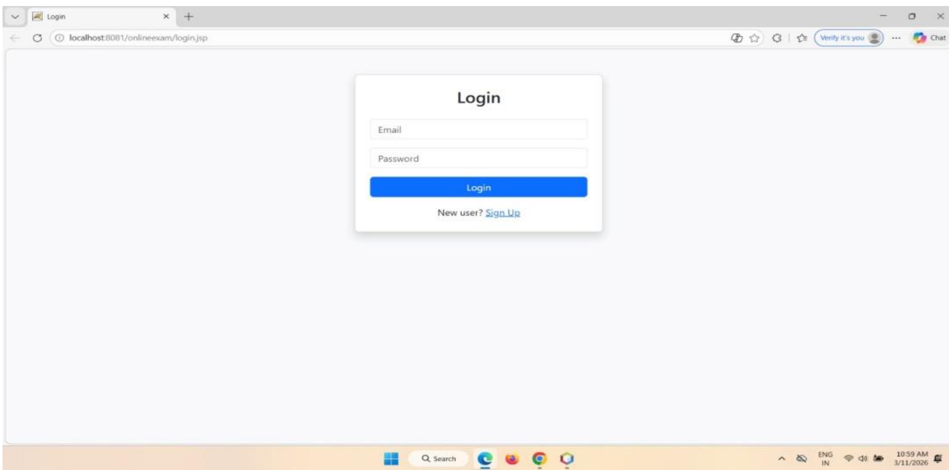
• Hero banner with exam tagline
• Feature cards (security, analytics, scalability)
• Informational sections
• Call-to-action buttons

The layout reflects a modern educational platform design.

5.4 Screenshots of the System

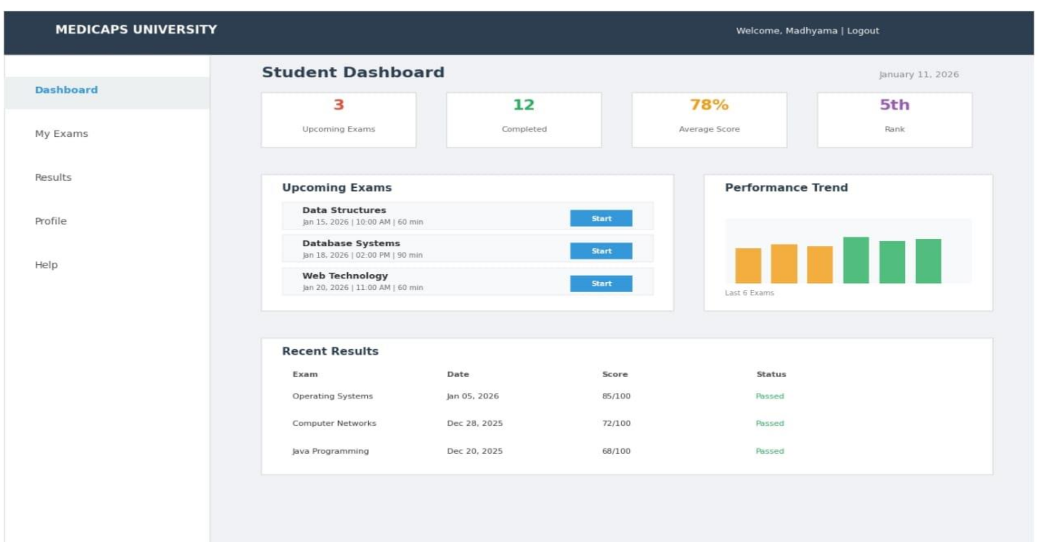
The system includes the following UI screens:

• Registration / Login page

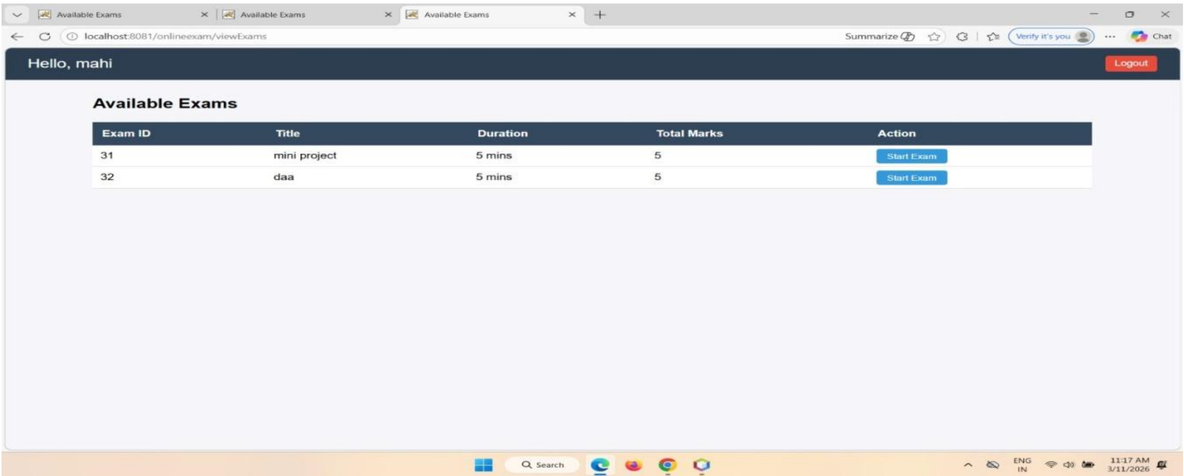


• Student dashboard

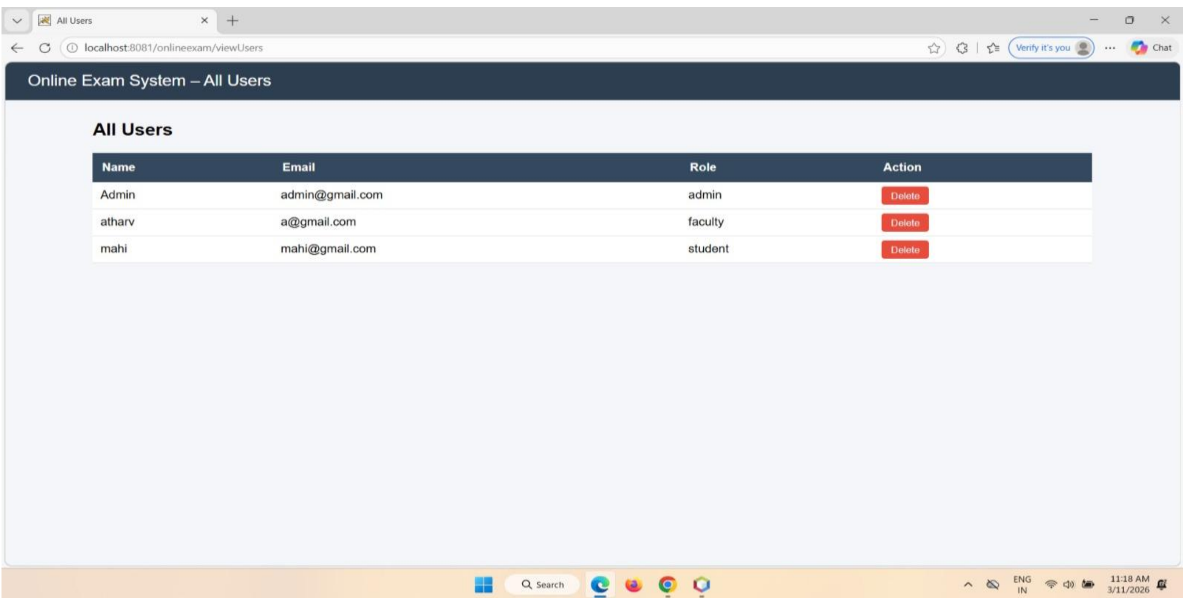
Figure 2: Student Dashboard Interface



- Examination interface



• Admin dashboard



• Result display screen

5.5 Services

The Online Examination System provides:

1. Online Examination Service

Conducts exams digitally with automated evaluation.

2. Secure Authentication Service

Ensures safe login and access control.

3. Automated Result Processing

Provides instant results with accuracy.

5.6 About

The **Online Examination System** is a web-based platform designed to simplify and automate the examination process.

It provides:

- Secure login and user management
- Efficient exam conduction
- Automatic result evaluation
- Scalable and user-friendly interface

The system improves efficiency, accuracy, and accessibility in modern education.

Chapter 6: Summary and Conclusions

The **Online Examination System** successfully provides a secure, efficient, and user-friendly platform for conducting examinations digitally. It is designed for educational institutions, students, and administrators to simplify the traditional examination process through automation.

The system is developed using a **client–server architecture**, where the frontend is built using **HTML, CSS, and JavaScript** to provide an interactive user interface, while the backend is implemented using **Java Servlets** for handling business logic and request processing. **MySQL** is used as the database for storing user data, exam details, questions, and results.

Unlike traditional paper-based examinations, the system automates the entire process including exam conduction, answer submission, and result evaluation. This reduces manual effort, minimizes errors, and improves overall efficiency.

The core functionality of the system follows a structured workflow:

User Login → Fetch Questions → Attempt Exam → Submit Exam → Evaluation → Result Display

This workflow ensures smooth operation, real-time processing, and instant feedback for users.

Security is an important aspect of the system. The application implements:

• Secure user authentication and login validation
• Session management to prevent unauthorized access
• Controlled access for students and administrators
• Database security for storing sensitive information

These features ensure data integrity, system reliability, and safe user interaction.

From a usability perspective, the system provides:

• A simple and responsive user interface
• Easy navigation for students and administrators
• Timer-based examination environment
• Instant result generation
• Efficient exam and question management

Overall, the **Online Examination System** improves accuracy, accessibility, and scalability in the examination process.

Chapter 7: Future Scope

The **Online Examination System** currently provides a secure and efficient platform for conducting online exams using web technologies and database integration. While it fulfills its core objectives, several future enhancements can improve its functionality and performance.

• Advanced Examination Features

Future improvements may include adaptive exams, random question generation, negative marking, and support for descriptive answers.

• AI-Based Proctoring

Integration of AI-based monitoring such as face detection, screen monitoring, and cheating detection can enhance exam security.

• Performance & Scalability

Cloud deployment, load balancing, and optimized database queries can improve performance for handling large numbers of users.

• Security Enhancements

Advanced encryption, two-factor authentication (2FA), and secure session handling can strengthen system security.

• User Experience Improvements

Mobile-friendly interface, dark mode, multilingual support, and accessibility features can enhance usability.

• Analytics & Reporting

Advanced analytics dashboards, performance tracking, and detailed reports can help institutions evaluate student performance.

• Integration Capabilities

Integration with Learning Management Systems (LMS) and third-party tools can extend system functionality.

• Research Expansion

Future work may include AI-based question generation, personalized exams, and predictive performance analysis.

Appendix

The appendix provides technical details and implementation overview of the system.

A. Code Structure Overview

The system follows a **client–server architecture**:

• Frontend

• HTML, CSS, JavaScript
• Responsive UI design
• Form validation

• Backend

• Java Servlets
• JDBC connectivity
• Business logic handling

• Database

• MySQL database
• Tables for users, exams, questions, results

B. Core Functional Modules

1. Authentication Module

• User registration and login
• Credential validation
• Session management

2. Exam Module

• Display questions
• Manage exam timing
• Submit answers

3. Result Module

• Automatic evaluation
• Score calculation
• Result display

4. Database Module

• User data storage
• Question and exam storage
• Result management

Sample Input and Output

Parameter	Example Value
Exam Type	MCQ Test
User Status	Authenticated
Request	Start Exam

Backend Processing Flow

1. User logs into the system
2. Exam questions are fetched from database
3. Student attempts questions
4. Answers are submitted
5. System evaluates responses
6. Result is calculated
7. Result is displayed

Sample Output

• Score displayed after submission
• Result page with marks and status
• Stored result in database

Testing

1. Unit Testing

• Login validation
• Question loading
• Answer evaluation

2. Integration Testing

• Frontend–backend communication
• Database connectivity

3. Manual Testing

• Login and exam flow
• UI responsiveness
• Error handling

4. Error Handling Tests

• Invalid login credentials
• Session timeout
• Database errors

5. Performance Testing

• Multiple user handling
• Exam submission speed

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